SG No. 181 D_6^5 $P6_422$ [hexagonal]

* plus set: $+[0, 0, 0]$
* Wyckoff site: `3a`, site symmetry: `222`

Table 1: Wyckoff bond: `3a@3a`, bond symmetry: `222`

No.	vector	center	mapping
1	$[X, 2X, 0]$	$[0, 0, 0]$	`[1,-4,-8,11]`
2	$[-2X, -X, 0]$	$[0, 0, \frac{1}{3}]$	`[2,-5,-7,10]`
3	$[X, -X, 0]$	$[0, 0, \frac{2}{3}]$	`[3,-6,-9,12]`

Table 2: Wyckoff bond: `3b@3a`, bond symmetry: `222`

No.	vector	center	mapping
1	$[X, 0, 0]$	$[0, 0, 0]$	`[1,-4,8,-11]`
2	$[0, X, 0]$	$[0, 0, \frac{1}{3}]$	`[2,-5,7,-10]`
3	$[-X, -X, 0]$	$[0, 0, \frac{2}{3}]$	`[3,-6,9,-12]`

Table 3: Wyckoff bond: `3c@3a`, bond symmetry: `222`

No.	vector	center	mapping
1	$[0, 0, Z]$	$[0, 0, 0]$	`[1,4,-8,-11]`
2	$[0, 0, Z]$	$[0, 0, \frac{1}{3}]$	`[2,5,-7,-10]`
3	$[0, 0, Z]$	$[0, 0, \frac{2}{3}]$	`[3,6,-9,-12]`

Table 4: Wyckoff bond: `6d@3a`, bond symmetry: `2..`

No.	vector	center	mapping
1	$[X, Y, 0]$	$[0, 0, 0]$	`[1,-4]`
2	$[-Y, X-Y, 0]$	$[0, 0, \frac{1}{3}]$	`[2,-5]`
3	$[-X+Y, -X, 0]$	$[0, 0, \frac{2}{3}]$	`[3,-6]`
4	$[Y, X, 0]$	$[0, 0, \frac{1}{3}]$	`[7,-10]`
5	$[X-Y, -Y, 0]$	$[0, 0, 0]$	`[8,-11]`
6	$[-X, -X+Y, 0]$	$[0, 0, \frac{2}{3}]$	`[9,-12]`

Table 5: Wyckoff bond: `6e@3a`, bond symmetry: `.2.`

No.	vector	center	mapping
1	$[X, 2X, Z]$	$[0, 0, 0]$	`[1,-8]`
2	$[-2X, -X, Z]$	$[0, 0, \frac{1}{3}]$	`[2,-7]`
3	$[X, -X, Z]$	$[0, 0, \frac{2}{3}]$	`[3,-9]`
4	$[-X, -2X, Z]$	$[0, 0, 0]$	`[4,-11]`
5	$[2X, X, Z]$	$[0, 0, \frac{1}{3}]$	`[5,-10]`
6	$[-X, X, Z]$	$[0, 0, \frac{2}{3}]$	`[6,-12]`

Table 6: Wyckoff bond: `6f@3a`, bond symmetry: `..2`

No.	vector	center	mapping
1	$[X, 0, Z]$	$[0, 0, 0]$	`[1,-11]`
2	$[0, X, Z]$	$\left[0, 0, \frac{1}{3}\right]$	`[2,-10]`
3	$[-X, -X, Z]$	$\left[0, 0, \frac{2}{3}\right]$	`[3,-12]`
4	$[-X, 0, Z]$	$[0, 0, 0]$	`[4,-8]`
5	$[0, -X, Z]$	$\left[0, 0, \frac{1}{3}\right]$	`[5,-7]`
6	$[X, X, Z]$	$\left[0, 0, \frac{2}{3}\right]$	`[6,-9]`

Table 7: Wyckoff bond: `12g@3a`, bond symmetry: `1`

No.	vector	center	mapping
1	$[X, Y, Z]$	$[0, 0, 0]$	`[1]`
2	$[-Y, X-Y, Z]$	$\left[0, 0, \frac{1}{3}\right]$	`[2]`
3	$[-X+Y, -X, Z]$	$\left[0, 0, \frac{2}{3}\right]$	`[3]`
4	$[-X, -Y, Z]$	$[0, 0, 0]$	`[4]`
5	$[Y, -X+Y, Z]$	$\left[0, 0, \frac{1}{3}\right]$	`[5]`
6	$[X-Y, X, Z]$	$\left[0, 0, \frac{2}{3}\right]$	`[6]`
7	$[Y, X, -Z]$	$\left[0, 0, \frac{1}{3}\right]$	`[7]`
8	$[X-Y, -Y, -Z]$	$[0, 0, 0]$	`[8]`
9	$[-X, -X+Y, -Z]$	$\left[0, 0, \frac{2}{3}\right]$	`[9]`
10	$[-Y, -X, -Z]$	$\left[0, 0, \frac{1}{3}\right]$	`[10]`
11	$[-X+Y, Y, -Z]$	$[0, 0, 0]$	`[11]`
12	$[X, X-Y, -Z]$	$\left[0, 0, \frac{2}{3}\right]$	`[12]`

* Wyckoff site: `3b`, site symmetry: `222`

Table 8: Wyckoff bond: `3a@3b`, bond symmetry: `222`

No.	vector	center	mapping
1	$[X, 2X, 0]$	$\left[0, 0, \frac{1}{2}\right]$	`[1,-4,-8,11]`
2	$[-2X, -X, 0]$	$\left[0, 0, \frac{5}{6}\right]$	`[2,-5,-7,10]`
3	$[X, -X, 0]$	$\left[0, 0, \frac{1}{6}\right]$	`[3,-6,-9,12]`

Table 9: Wyckoff bond: `3b@3b`, bond symmetry: `222`

No.	vector	center	mapping
1	$[X, 0, 0]$	$\left[0, 0, \frac{1}{2}\right]$	`[1,-4,8,-11]`
2	$[0, X, 0]$	$\left[0, 0, \frac{5}{6}\right]$	`[2,-5,7,-10]`
3	$[-X, -X, 0]$	$\left[0, 0, \frac{1}{6}\right]$	`[3,-6,9,-12]`

Table 10: Wyckoff bond: `3c@3b`, bond symmetry: `222`

No.	vector	center	mapping
1	$[0, 0, Z]$	$\left[0, 0, \frac{1}{2}\right]$	`[1,4,-8,-11]`

continued ...

Table 10

No.	vector	center	mapping
2	$[0,\ 0,\ Z]$	$\left[0,\ 0,\ \frac{5}{6}\right]$	`[2,5,-7,-10]`
3	$[0,\ 0,\ Z]$	$\left[0,\ 0,\ \frac{1}{6}\right]$	`[3,6,-9,-12]`

Table 11: Wyckoff bond: `6d@3b`, bond symmetry: `2..`

No.	vector	center	mapping
1	$[X,\ Y,\ 0]$	$\left[0,\ 0,\ \frac{1}{2}\right]$	`[1,-4]`
2	$[-Y,\ X-Y,\ 0]$	$\left[0,\ 0,\ \frac{5}{6}\right]$	`[2,-5]`
3	$[-X+Y,\ -X,\ 0]$	$\left[0,\ 0,\ \frac{1}{6}\right]$	`[3,-6]`
4	$[Y,\ X,\ 0]$	$\left[0,\ 0,\ \frac{5}{6}\right]$	`[7,-10]`
5	$[X-Y,\ -Y,\ 0]$	$\left[0,\ 0,\ \frac{1}{2}\right]$	`[8,-11]`
6	$[-X,\ -X+Y,\ 0]$	$\left[0,\ 0,\ \frac{1}{6}\right]$	`[9,-12]`

Table 12: Wyckoff bond: `6e@3b`, bond symmetry: `.2.`

No.	vector	center	mapping
1	$[X,\ 2X,\ Z]$	$\left[0,\ 0,\ \frac{1}{2}\right]$	`[1,-8]`
2	$[-2X,\ -X,\ Z]$	$\left[0,\ 0,\ \frac{5}{6}\right]$	`[2,-7]`
3	$[X,\ -X,\ Z]$	$\left[0,\ 0,\ \frac{1}{6}\right]$	`[3,-9]`
4	$[-X,\ -2X,\ Z]$	$\left[0,\ 0,\ \frac{1}{2}\right]$	`[4,-11]`
5	$[2X,\ X,\ Z]$	$\left[0,\ 0,\ \frac{5}{6}\right]$	`[5,-10]`
6	$[-X,\ X,\ Z]$	$\left[0,\ 0,\ \frac{1}{6}\right]$	`[6,-12]`

Table 13: Wyckoff bond: `6f@3b`, bond symmetry: `..2`

No.	vector	center	mapping
1	$[X,\ 0,\ Z]$	$\left[0,\ 0,\ \frac{1}{2}\right]$	`[1,-11]`
2	$[0,\ X,\ Z]$	$\left[0,\ 0,\ \frac{5}{6}\right]$	`[2,-10]`
3	$[-X,\ -X,\ Z]$	$\left[0,\ 0,\ \frac{1}{6}\right]$	`[3,-12]`
4	$[-X,\ 0,\ Z]$	$\left[0,\ 0,\ \frac{1}{2}\right]$	`[4,-8]`
5	$[0,\ -X,\ Z]$	$\left[0,\ 0,\ \frac{5}{6}\right]$	`[5,-7]`
6	$[X,\ X,\ Z]$	$\left[0,\ 0,\ \frac{1}{6}\right]$	`[6,-9]`

Table 14: Wyckoff bond: `12g@3b`, bond symmetry: `1`

No.	vector	center	mapping
1	$[X,\ Y,\ Z]$	$\left[0,\ 0,\ \frac{1}{2}\right]$	`[1]`
2	$[-Y,\ X-Y,\ Z]$	$\left[0,\ 0,\ \frac{5}{6}\right]$	`[2]`
3	$[-X+Y,\ -X,\ Z]$	$\left[0,\ 0,\ \frac{1}{6}\right]$	`[3]`
4	$[-X,\ -Y,\ Z]$	$\left[0,\ 0,\ \frac{1}{2}\right]$	`[4]`
5	$[Y,\ -X+Y,\ Z]$	$\left[0,\ 0,\ \frac{5}{6}\right]$	`[5]`
6	$[X-Y,\ X,\ Z]$	$\left[0,\ 0,\ \frac{1}{6}\right]$	`[6]`

continued ...

Table 14

No.	vector	center	mapping
7	$[Y, X, -Z]$	$[0, 0, \frac{5}{6}]$	`[7]`
8	$[X-Y, -Y, -Z]$	$[0, 0, \frac{1}{2}]$	`[8]`
9	$[-X, -X+Y, -Z]$	$[0, 0, \frac{1}{6}]$	`[9]`
10	$[-Y, -X, -Z]$	$[0, 0, \frac{5}{6}]$	`[10]`
11	$[-X+Y, Y, -Z]$	$[0, 0, \frac{1}{2}]$	`[11]`
12	$[X, X-Y, -Z]$	$[0, 0, \frac{1}{6}]$	`[12]`

* Wyckoff site: `3c`, site symmetry: `222`

Table 15: Wyckoff bond: `3a@3c`, bond symmetry: `222`

No.	vector	center	mapping
1	$[X, 2X, 0]$	$[\frac{1}{2}, 0, 0]$	`[1,-4,-8,11]`
2	$[-2X, -X, 0]$	$[0, \frac{1}{2}, \frac{1}{3}]$	`[2,-5,-7,10]`
3	$[X, -X, 0]$	$[\frac{1}{2}, \frac{1}{2}, \frac{2}{3}]$	`[3,-6,-9,12]`

Table 16: Wyckoff bond: `3b@3c`, bond symmetry: `222`

No.	vector	center	mapping
1	$[X, 0, 0]$	$[\frac{1}{2}, 0, 0]$	`[1,-4,8,-11]`
2	$[0, X, 0]$	$[0, \frac{1}{2}, \frac{1}{3}]$	`[2,-5,7,-10]`
3	$[-X, -X, 0]$	$[\frac{1}{2}, \frac{1}{2}, \frac{2}{3}]$	`[3,-6,9,-12]`

Table 17: Wyckoff bond: `3c@3c`, bond symmetry: `222`

No.	vector	center	mapping
1	$[0, 0, Z]$	$[\frac{1}{2}, 0, 0]$	`[1,4,-8,-11]`
2	$[0, 0, Z]$	$[0, \frac{1}{2}, \frac{1}{3}]$	`[2,5,-7,-10]`
3	$[0, 0, Z]$	$[\frac{1}{2}, \frac{1}{2}, \frac{2}{3}]$	`[3,6,-9,-12]`

Table 18: Wyckoff bond: `6d@3c`, bond symmetry: `2..`

No.	vector	center	mapping
1	$[X, Y, 0]$	$[\frac{1}{2}, 0, 0]$	`[1,-4]`
2	$[-Y, X-Y, 0]$	$[0, \frac{1}{2}, \frac{1}{3}]$	`[2,-5]`
3	$[-X+Y, -X, 0]$	$[\frac{1}{2}, \frac{1}{2}, \frac{2}{3}]$	`[3,-6]`
4	$[Y, X, 0]$	$[0, \frac{1}{2}, \frac{1}{3}]$	`[7,-10]`
5	$[X-Y, -Y, 0]$	$[\frac{1}{2}, 0, 0]$	`[8,-11]`
6	$[-X, -X+Y, 0]$	$[\frac{1}{2}, \frac{1}{2}, \frac{2}{3}]$	`[9,-12]`

Table 19: Wyckoff bond: `6e@3c`, bond symmetry: `.2.`

No.	vector	center	mapping
1	$[X, 2X, Z]$	$\left[\frac{1}{2}, 0, 0\right]$	`[1,-8]`
2	$[-2X, -X, Z]$	$\left[0, \frac{1}{2}, \frac{1}{3}\right]$	`[2,-7]`
3	$[X, -X, Z]$	$\left[\frac{1}{2}, \frac{1}{2}, \frac{2}{3}\right]$	`[3,-9]`
4	$[-X, -2X, Z]$	$\left[\frac{1}{2}, 0, 0\right]$	`[4,-11]`
5	$[2X, X, Z]$	$\left[0, \frac{1}{2}, \frac{1}{3}\right]$	`[5,-10]`
6	$[-X, X, Z]$	$\left[\frac{1}{2}, \frac{1}{2}, \frac{2}{3}\right]$	`[6,-12]`

Table 20: Wyckoff bond: `6f@3c`, bond symmetry: `..2`

No.	vector	center	mapping
1	$[X, 0, Z]$	$\left[\frac{1}{2}, 0, 0\right]$	`[1,-11]`
2	$[0, X, Z]$	$\left[0, \frac{1}{2}, \frac{1}{3}\right]$	`[2,-10]`
3	$[-X, -X, Z]$	$\left[\frac{1}{2}, \frac{1}{2}, \frac{2}{3}\right]$	`[3,-12]`
4	$[-X, 0, Z]$	$\left[\frac{1}{2}, 0, 0\right]$	`[4,-8]`
5	$[0, -X, Z]$	$\left[0, \frac{1}{2}, \frac{1}{3}\right]$	`[5,-7]`
6	$[X, X, Z]$	$\left[\frac{1}{2}, \frac{1}{2}, \frac{2}{3}\right]$	`[6,-9]`

Table 21: Wyckoff bond: `12g@3c`, bond symmetry: `1`

No.	vector	center	mapping
1	$[X, Y, Z]$	$\left[\frac{1}{2}, 0, 0\right]$	`[1]`
2	$[-Y, X-Y, Z]$	$\left[0, \frac{1}{2}, \frac{1}{3}\right]$	`[2]`
3	$[-X+Y, -X, Z]$	$\left[\frac{1}{2}, \frac{1}{2}, \frac{2}{3}\right]$	`[3]`
4	$[-X, -Y, Z]$	$\left[\frac{1}{2}, 0, 0\right]$	`[4]`
5	$[Y, -X+Y, Z]$	$\left[0, \frac{1}{2}, \frac{1}{3}\right]$	`[5]`
6	$[X-Y, X, Z]$	$\left[\frac{1}{2}, \frac{1}{2}, \frac{2}{3}\right]$	`[6]`
7	$[Y, X, -Z]$	$\left[0, \frac{1}{2}, \frac{1}{3}\right]$	`[7]`
8	$[X-Y, -Y, -Z]$	$\left[\frac{1}{2}, 0, 0\right]$	`[8]`
9	$[-X, -X+Y, -Z]$	$\left[\frac{1}{2}, \frac{1}{2}, \frac{2}{3}\right]$	`[9]`
10	$[-Y, -X, -Z]$	$\left[0, \frac{1}{2}, \frac{1}{3}\right]$	`[10]`
11	$[-X+Y, Y, -Z]$	$\left[\frac{1}{2}, 0, 0\right]$	`[11]`
12	$[X, X-Y, -Z]$	$\left[\frac{1}{2}, \frac{1}{2}, \frac{2}{3}\right]$	`[12]`

* Wyckoff site: `3d`, site symmetry: `222`

Table 22: Wyckoff bond: `3a@3d`, bond symmetry: `222`

No.	vector	center	mapping
1	$[X, 2X, 0]$	$\left[\frac{1}{2}, 0, \frac{1}{2}\right]$	`[1,-4,-8,11]`
2	$[-2X, -X, 0]$	$\left[0, \frac{1}{2}, \frac{5}{6}\right]$	`[2,-5,-7,10]`
3	$[X, -X, 0]$	$\left[\frac{1}{2}, \frac{1}{2}, \frac{1}{6}\right]$	`[3,-6,-9,12]`

Table 23: Wyckoff bond: `3b@3d`, bond symmetry: `222`

No.	vector	center	mapping
1	$[X, 0, 0]$	$\left[\frac{1}{2}, 0, \frac{1}{2}\right]$	`[1,-4,8,-11]`
2	$[0, X, 0]$	$\left[0, \frac{1}{2}, \frac{5}{6}\right]$	`[2,-5,7,-10]`
3	$[-X, -X, 0]$	$\left[\frac{1}{2}, \frac{1}{2}, \frac{1}{6}\right]$	`[3,-6,9,-12]`

Table 24: Wyckoff bond: `3c@3d`, bond symmetry: `222`

No.	vector	center	mapping
1	$[0, 0, Z]$	$\left[\frac{1}{2}, 0, \frac{1}{2}\right]$	`[1,4,-8,-11]`
2	$[0, 0, Z]$	$\left[0, \frac{1}{2}, \frac{5}{6}\right]$	`[2,5,-7,-10]`
3	$[0, 0, Z]$	$\left[\frac{1}{2}, \frac{1}{2}, \frac{1}{6}\right]$	`[3,6,-9,-12]`

Table 25: Wyckoff bond: `6d@3d`, bond symmetry: `2..`

No.	vector	center	mapping
1	$[X, Y, 0]$	$\left[\frac{1}{2}, 0, \frac{1}{2}\right]$	`[1,-4]`
2	$[-Y, X-Y, 0]$	$\left[0, \frac{1}{2}, \frac{5}{6}\right]$	`[2,-5]`
3	$[-X+Y, -X, 0]$	$\left[\frac{1}{2}, \frac{1}{2}, \frac{1}{6}\right]$	`[3,-6]`
4	$[Y, X, 0]$	$\left[0, \frac{1}{2}, \frac{5}{6}\right]$	`[7,-10]`
5	$[X-Y, -Y, 0]$	$\left[\frac{1}{2}, 0, \frac{1}{2}\right]$	`[8,-11]`
6	$[-X, -X+Y, 0]$	$\left[\frac{1}{2}, \frac{1}{2}, \frac{1}{6}\right]$	`[9,-12]`

Table 26: Wyckoff bond: `6e@3d`, bond symmetry: `.2.`

No.	vector	center	mapping
1	$[X, 2X, Z]$	$\left[\frac{1}{2}, 0, \frac{1}{2}\right]$	`[1,-8]`
2	$[-2X, -X, Z]$	$\left[0, \frac{1}{2}, \frac{5}{6}\right]$	`[2,-7]`
3	$[X, -X, Z]$	$\left[\frac{1}{2}, \frac{1}{2}, \frac{1}{6}\right]$	`[3,-9]`
4	$[-X, -2X, Z]$	$\left[\frac{1}{2}, 0, \frac{1}{2}\right]$	`[4,-11]`
5	$[2X, X, Z]$	$\left[0, \frac{1}{2}, \frac{5}{6}\right]$	`[5,-10]`
6	$[-X, X, Z]$	$\left[\frac{1}{2}, \frac{1}{2}, \frac{1}{6}\right]$	`[6,-12]`

Table 27: Wyckoff bond: `6f@3d`, bond symmetry: `..2`

No.	vector	center	mapping
1	$[X, 0, Z]$	$\left[\frac{1}{2}, 0, \frac{1}{2}\right]$	`[1,-11]`
2	$[0, X, Z]$	$\left[0, \frac{1}{2}, \frac{5}{6}\right]$	`[2,-10]`
3	$[-X, -X, Z]$	$\left[\frac{1}{2}, \frac{1}{2}, \frac{1}{6}\right]$	`[3,-12]`
4	$[-X, 0, Z]$	$\left[\frac{1}{2}, 0, \frac{1}{2}\right]$	`[4,-8]`
5	$[0, -X, Z]$	$\left[0, \frac{1}{2}, \frac{5}{6}\right]$	`[5,-7]`
6	$[X, X, Z]$	$\left[\frac{1}{2}, \frac{1}{2}, \frac{1}{6}\right]$	`[6,-9]`

Table 28: Wyckoff bond: `12g@3d`, bond symmetry: `1`

No.	vector	center	mapping
1	$[X,\ Y,\ Z]$	$[\frac{1}{2},\ 0,\ \frac{1}{2}]$	`[1]`
2	$[-Y,\ X-Y,\ Z]$	$[0,\ \frac{1}{2},\ \frac{5}{6}]$	`[2]`
3	$[-X+Y,\ -X,\ Z]$	$[\frac{1}{2},\ \frac{1}{2},\ \frac{1}{6}]$	`[3]`
4	$[-X,\ -Y,\ Z]$	$[\frac{1}{2},\ 0,\ \frac{1}{2}]$	`[4]`
5	$[Y,\ -X+Y,\ Z]$	$[0,\ \frac{1}{2},\ \frac{5}{6}]$	`[5]`
6	$[X-Y,\ X,\ Z]$	$[\frac{1}{2},\ \frac{1}{2},\ \frac{1}{6}]$	`[6]`
7	$[Y,\ X,\ -Z]$	$[0,\ \frac{1}{2},\ \frac{5}{6}]$	`[7]`
8	$[X-Y,\ -Y,\ -Z]$	$[\frac{1}{2},\ 0,\ \frac{1}{2}]$	`[8]`
9	$[-X,\ -X+Y,\ -Z]$	$[\frac{1}{2},\ \frac{1}{2},\ \frac{1}{6}]$	`[9]`
10	$[-Y,\ -X,\ -Z]$	$[0,\ \frac{1}{2},\ \frac{5}{6}]$	`[10]`
11	$[-X+Y,\ Y,\ -Z]$	$[\frac{1}{2},\ 0,\ \frac{1}{2}]$	`[11]`
12	$[X,\ X-Y,\ -Z]$	$[\frac{1}{2},\ \frac{1}{2},\ \frac{1}{6}]$	`[12]`

* Wyckoff site: `6e`, site symmetry: `2..`

Table 29: Wyckoff bond: `6a@6e`, bond symmetry: `2..`

No.	vector	center	mapping
1	$[X,\ Y,\ 0]$	$[0,\ 0,\ z]$	`[1,-4]`
2	$[-Y,\ X-Y,\ 0]$	$[0,\ 0,\ z+\frac{1}{3}]$	`[2,-5]`
3	$[-X+Y,\ -X,\ 0]$	$[0,\ 0,\ z+\frac{2}{3}]$	`[3,-6]`
4	$[Y,\ X,\ 0]$	$[0,\ 0,\ \frac{1}{3}-z]$	`[7,-10]`
5	$[X-Y,\ -Y,\ 0]$	$[0,\ 0,\ -z]$	`[8,-11]`
6	$[-X,\ -X+Y,\ 0]$	$[0,\ 0,\ \frac{2}{3}-z]$	`[9,-12]`

Table 30: Wyckoff bond: `6b@6e`, bond symmetry: `2..`

No.	vector	center	mapping
1	$[0,\ 0,\ Z]$	$[0,\ 0,\ z]$	`[1,4]`
2	$[0,\ 0,\ Z]$	$[0,\ 0,\ z+\frac{1}{3}]$	`[2,5]`
3	$[0,\ 0,\ Z]$	$[0,\ 0,\ z+\frac{2}{3}]$	`[3,6]`
4	$[0,\ 0,\ -Z]$	$[0,\ 0,\ \frac{1}{3}-z]$	`[7,10]`
5	$[0,\ 0,\ -Z]$	$[0,\ 0,\ -z]$	`[8,11]`
6	$[0,\ 0,\ -Z]$	$[0,\ 0,\ \frac{2}{3}-z]$	`[9,12]`

Table 31: Wyckoff bond: `12c@6e`, bond symmetry: `1`

No.	vector	center	mapping
1	$[X,\ Y,\ Z]$	$[0,\ 0,\ z]$	`[1]`
2	$[-Y,\ X-Y,\ Z]$	$[0,\ 0,\ z+\frac{1}{3}]$	`[2]`
3	$[-X+Y,\ -X,\ Z]$	$[0,\ 0,\ z+\frac{2}{3}]$	`[3]`
4	$[-X,\ -Y,\ Z]$	$[0,\ 0,\ z]$	`[4]`
5	$[Y,\ -X+Y,\ Z]$	$[0,\ 0,\ z+\frac{1}{3}]$	`[5]`
6	$[X-Y,\ X,\ Z]$	$[0,\ 0,\ z+\frac{2}{3}]$	`[6]`

continued ...

Table 31

No.	vector	center	mapping
7	$[Y,\ X,\ -Z]$	$\left[0,\ 0,\ \frac{1}{3}-z\right]$	`[7]`
8	$[X-Y,\ -Y,\ -Z]$	$[0,\ 0,\ -z]$	`[8]`
9	$[-X,\ -X+Y,\ -Z]$	$\left[0,\ 0,\ \frac{2}{3}-z\right]$	`[9]`
10	$[-Y,\ -X,\ -Z]$	$\left[0,\ 0,\ \frac{1}{3}-z\right]$	`[10]`
11	$[-X+Y,\ Y,\ -Z]$	$[0,\ 0,\ -z]$	`[11]`
12	$[X,\ X-Y,\ -Z]$	$\left[0,\ 0,\ \frac{2}{3}-z\right]$	`[12]`

* Wyckoff site: `6f`, site symmetry: `2..`

Table 32: Wyckoff bond: `6a@6f`, bond symmetry: `2..`

No.	vector	center	mapping
1	$[X,\ Y,\ 0]$	$\left[\frac{1}{2},\ 0,\ z\right]$	`[1,-4]`
2	$[-Y,\ X-Y,\ 0]$	$\left[0,\ \frac{1}{2},\ z+\frac{1}{3}\right]$	`[2,-5]`
3	$[-X+Y,\ -X,\ 0]$	$\left[\frac{1}{2},\ \frac{1}{2},\ z+\frac{2}{3}\right]$	`[3,-6]`
4	$[Y,\ X,\ 0]$	$\left[0,\ \frac{1}{2},\ \frac{1}{3}-z\right]$	`[7,-10]`
5	$[X-Y,\ -Y,\ 0]$	$\left[\frac{1}{2},\ 0,\ -z\right]$	`[8,-11]`
6	$[-X,\ -X+Y,\ 0]$	$\left[\frac{1}{2},\ \frac{1}{2},\ \frac{2}{3}-z\right]$	`[9,-12]`

Table 33: Wyckoff bond: `6b@6f`, bond symmetry: `2..`

No.	vector	center	mapping
1	$[0,\ 0,\ Z]$	$\left[\frac{1}{2},\ 0,\ z\right]$	`[1,4]`
2	$[0,\ 0,\ Z]$	$\left[0,\ \frac{1}{2},\ z+\frac{1}{3}\right]$	`[2,5]`
3	$[0,\ 0,\ Z]$	$\left[\frac{1}{2},\ \frac{1}{2},\ z+\frac{2}{3}\right]$	`[3,6]`
4	$[0,\ 0,\ -Z]$	$\left[0,\ \frac{1}{2},\ \frac{1}{3}-z\right]$	`[7,10]`
5	$[0,\ 0,\ -Z]$	$\left[\frac{1}{2},\ 0,\ -z\right]$	`[8,11]`
6	$[0,\ 0,\ -Z]$	$\left[\frac{1}{2},\ \frac{1}{2},\ \frac{2}{3}-z\right]$	`[9,12]`

Table 34: Wyckoff bond: `12c@6f`, bond symmetry: `1`

No.	vector	center	mapping
1	$[X,\ Y,\ Z]$	$\left[\frac{1}{2},\ 0,\ z\right]$	`[1]`
2	$[-Y,\ X-Y,\ Z]$	$\left[0,\ \frac{1}{2},\ z+\frac{1}{3}\right]$	`[2]`
3	$[-X+Y,\ -X,\ Z]$	$\left[\frac{1}{2},\ \frac{1}{2},\ z+\frac{2}{3}\right]$	`[3]`
4	$[-X,\ -Y,\ Z]$	$\left[\frac{1}{2},\ 0,\ z\right]$	`[4]`
5	$[Y,\ -X+Y,\ Z]$	$\left[0,\ \frac{1}{2},\ z+\frac{1}{3}\right]$	`[5]`
6	$[X-Y,\ X,\ Z]$	$\left[\frac{1}{2},\ \frac{1}{2},\ z+\frac{2}{3}\right]$	`[6]`
7	$[Y,\ X,\ -Z]$	$\left[0,\ \frac{1}{2},\ \frac{1}{3}-z\right]$	`[7]`
8	$[X-Y,\ -Y,\ -Z]$	$\left[\frac{1}{2},\ 0,\ -z\right]$	`[8]`
9	$[-X,\ -X+Y,\ -Z]$	$\left[\frac{1}{2},\ \frac{1}{2},\ \frac{2}{3}-z\right]$	`[9]`
10	$[-Y,\ -X,\ -Z]$	$\left[0,\ \frac{1}{2},\ \frac{1}{3}-z\right]$	`[10]`
11	$[-X+Y,\ Y,\ -Z]$	$\left[\frac{1}{2},\ 0,\ -z\right]$	`[11]`
12	$[X,\ X-Y,\ -Z]$	$\left[\frac{1}{2},\ \frac{1}{2},\ \frac{2}{3}-z\right]$	`[12]`

* Wyckoff site: `6g`, site symmetry: `.2.`

Table 35: Wyckoff bond: `6a@6g`, bond symmetry: `.2.`

No.	vector	center	mapping
1	$[X,\ 2X,\ Z]$	$[x,\ 0,\ 0]$	`[1,-8]`
2	$[-2X,\ -X,\ Z]$	$\left[0,\ x,\ \frac{1}{3}\right]$	`[2,-7]`
3	$[X,\ -X,\ Z]$	$\left[-x,\ -x,\ \frac{2}{3}\right]$	`[3,-9]`
4	$[-X,\ -2X,\ Z]$	$[-x,\ 0,\ 0]$	`[4,-11]`
5	$[2X,\ X,\ Z]$	$\left[0,\ -x,\ \frac{1}{3}\right]$	`[5,-10]`
6	$[-X,\ X,\ Z]$	$\left[x,\ x,\ \frac{2}{3}\right]$	`[6,-12]`

Table 36: Wyckoff bond: `6b@6g`, bond symmetry: `.2.`

No.	vector	center	mapping
1	$[X,\ 0,\ 0]$	$[x,\ 0,\ 0]$	`[1,8]`
2	$[0,\ X,\ 0]$	$\left[0,\ x,\ \frac{1}{3}\right]$	`[2,7]`
3	$[-X,\ -X,\ 0]$	$\left[-x,\ -x,\ \frac{2}{3}\right]$	`[3,9]`
4	$[-X,\ 0,\ 0]$	$[-x,\ 0,\ 0]$	`[4,11]`
5	$[0,\ -X,\ 0]$	$\left[0,\ -x,\ \frac{1}{3}\right]$	`[5,10]`
6	$[X,\ X,\ 0]$	$\left[x,\ x,\ \frac{2}{3}\right]$	`[6,12]`

Table 37: Wyckoff bond: `12c@6g`, bond symmetry: `1`

No.	vector	center	mapping
1	$[X,\ Y,\ Z]$	$[x,\ 0,\ 0]$	`[1]`
2	$[-Y,\ X-Y,\ Z]$	$\left[0,\ x,\ \frac{1}{3}\right]$	`[2]`
3	$[-X+Y,\ -X,\ Z]$	$\left[-x,\ -x,\ \frac{2}{3}\right]$	`[3]`
4	$[-X,\ -Y,\ Z]$	$[-x,\ 0,\ 0]$	`[4]`
5	$[Y,\ -X+Y,\ Z]$	$\left[0,\ -x,\ \frac{1}{3}\right]$	`[5]`
6	$[X-Y,\ X,\ Z]$	$\left[x,\ x,\ \frac{2}{3}\right]$	`[6]`
7	$[Y,\ X,\ -Z]$	$\left[0,\ x,\ \frac{1}{3}\right]$	`[7]`
8	$[X-Y,\ -Y,\ -Z]$	$[x,\ 0,\ 0]$	`[8]`
9	$[-X,\ -X+Y,\ -Z]$	$\left[-x,\ -x,\ \frac{2}{3}\right]$	`[9]`
10	$[-Y,\ -X,\ -Z]$	$\left[0,\ -x,\ \frac{1}{3}\right]$	`[10]`
11	$[-X+Y,\ Y,\ -Z]$	$[-x,\ 0,\ 0]$	`[11]`
12	$[X,\ X-Y,\ -Z]$	$\left[x,\ x,\ \frac{2}{3}\right]$	`[12]`

* Wyckoff site: `6h`, site symmetry: `.2.`

Table 38: Wyckoff bond: `6a@6h`, bond symmetry: `.2.`

No.	vector	center	mapping
1	$[X,\ 2X,\ Z]$	$\left[x,\ 0,\ \frac{1}{2}\right]$	`[1,-8]`
2	$[-2X,\ -X,\ Z]$	$\left[0,\ x,\ \frac{5}{6}\right]$	`[2,-7]`
3	$[X,\ -X,\ Z]$	$\left[-x,\ -x,\ \frac{1}{6}\right]$	`[3,-9]`
4	$[-X,\ -2X,\ Z]$	$\left[-x,\ 0,\ \frac{1}{2}\right]$	`[4,-11]`
5	$[2X,\ X,\ Z]$	$\left[0,\ -x,\ \frac{5}{6}\right]$	`[5,-10]`

continued ...

Table 38

No.	vector	center	mapping
6	$[-X,\, X,\, Z]$	$\left[x,\, x,\, \frac{1}{6}\right]$	`[6,-12]`

Table 39: Wyckoff bond: `6b@6h`, bond symmetry: `.2.`

No.	vector	center	mapping
1	$[X,\, 0,\, 0]$	$\left[x,\, 0,\, \frac{1}{2}\right]$	`[1,8]`
2	$[0,\, X,\, 0]$	$\left[0,\, x,\, \frac{5}{6}\right]$	`[2,7]`
3	$[-X,\, -X,\, 0]$	$\left[-x,\, -x,\, \frac{1}{6}\right]$	`[3,9]`
4	$[-X,\, 0,\, 0]$	$\left[-x,\, 0,\, \frac{1}{2}\right]$	`[4,11]`
5	$[0,\, -X,\, 0]$	$\left[0,\, -x,\, \frac{5}{6}\right]$	`[5,10]`
6	$[X,\, X,\, 0]$	$\left[x,\, x,\, \frac{1}{6}\right]$	`[6,12]`

Table 40: Wyckoff bond: `12c@6h`, bond symmetry: `1`

No.	vector	center	mapping
1	$[X,\, Y,\, Z]$	$\left[x,\, 0,\, \frac{1}{2}\right]$	`[1]`
2	$[-Y,\, X-Y,\, Z]$	$\left[0,\, x,\, \frac{5}{6}\right]$	`[2]`
3	$[-X+Y,\, -X,\, Z]$	$\left[-x,\, -x,\, \frac{1}{6}\right]$	`[3]`
4	$[-X,\, -Y,\, Z]$	$\left[-x,\, 0,\, \frac{1}{2}\right]$	`[4]`
5	$[Y,\, -X+Y,\, Z]$	$\left[0,\, -x,\, \frac{5}{6}\right]$	`[5]`
6	$[X-Y,\, X,\, Z]$	$\left[x,\, x,\, \frac{1}{6}\right]$	`[6]`
7	$[Y,\, X,\, -Z]$	$\left[0,\, x,\, \frac{5}{6}\right]$	`[7]`
8	$[X-Y,\, -Y,\, -Z]$	$\left[x,\, 0,\, \frac{1}{2}\right]$	`[8]`
9	$[-X,\, -X+Y,\, -Z]$	$\left[-x,\, -x,\, \frac{1}{6}\right]$	`[9]`
10	$[-Y,\, -X,\, -Z]$	$\left[0,\, -x,\, \frac{5}{6}\right]$	`[10]`
11	$[-X+Y,\, Y,\, -Z]$	$\left[-x,\, 0,\, \frac{1}{2}\right]$	`[11]`
12	$[X,\, X-Y,\, -Z]$	$\left[x,\, x,\, \frac{1}{6}\right]$	`[12]`

* Wyckoff site: `6i`, site symmetry: `..2`

Table 41: Wyckoff bond: `6a@6i`, bond symmetry: `..2`

No.	vector	center	mapping
1	$[X,\, 0,\, Z]$	$[x,\, 2x,\, 0]$	`[1,-11]`
2	$[0,\, X,\, Z]$	$\left[-2x,\, -x,\, \frac{1}{3}\right]$	`[2,-10]`
3	$[-X,\, -X,\, Z]$	$\left[x,\, -x,\, \frac{2}{3}\right]$	`[3,-12]`
4	$[-X,\, 0,\, Z]$	$[-x,\, -2x,\, 0]$	`[4,-8]`
5	$[0,\, -X,\, Z]$	$\left[2x,\, x,\, \frac{1}{3}\right]$	`[5,-7]`
6	$[X,\, X,\, Z]$	$\left[-x,\, x,\, \frac{2}{3}\right]$	`[6,-9]`

Table 42: Wyckoff bond: `6b@6i`, bond symmetry: `..2`

No.	vector	center	mapping
1	$[X,\ 2X,\ 0]$	$[x,\ 2x,\ 0]$	`[1,11]`
2	$[-2X,\ -X,\ 0]$	$\left[-2x,\ -x,\ \frac{1}{3}\right]$	`[2,10]`
3	$[X,\ -X,\ 0]$	$\left[x,\ -x,\ \frac{2}{3}\right]$	`[3,12]`
4	$[-X,\ -2X,\ 0]$	$[-x,\ -2x,\ 0]$	`[4,8]`
5	$[2X,\ X,\ 0]$	$\left[2x,\ x,\ \frac{1}{3}\right]$	`[5,7]`
6	$[-X,\ X,\ 0]$	$\left[-x,\ x,\ \frac{2}{3}\right]$	`[6,9]`

Table 43: Wyckoff bond: `12c@6i`, bond symmetry: `1`

No.	vector	center	mapping
1	$[X,\ Y,\ Z]$	$[x,\ 2x,\ 0]$	`[1]`
2	$[-Y,\ X-Y,\ Z]$	$\left[-2x,\ -x,\ \frac{1}{3}\right]$	`[2]`
3	$[-X+Y,\ -X,\ Z]$	$\left[x,\ -x,\ \frac{2}{3}\right]$	`[3]`
4	$[-X,\ -Y,\ Z]$	$[-x,\ -2x,\ 0]$	`[4]`
5	$[Y,\ -X+Y,\ Z]$	$\left[2x,\ x,\ \frac{1}{3}\right]$	`[5]`
6	$[X-Y,\ X,\ Z]$	$\left[-x,\ x,\ \frac{2}{3}\right]$	`[6]`
7	$[Y,\ X,\ -Z]$	$\left[2x,\ x,\ \frac{1}{3}\right]$	`[7]`
8	$[X-Y,\ -Y,\ -Z]$	$[-x,\ -2x,\ 0]$	`[8]`
9	$[-X,\ -X+Y,\ -Z]$	$\left[-x,\ x,\ \frac{2}{3}\right]$	`[9]`
10	$[-Y,\ -X,\ -Z]$	$\left[-2x,\ -x,\ \frac{1}{3}\right]$	`[10]`
11	$[-X+Y,\ Y,\ -Z]$	$[x,\ 2x,\ 0]$	`[11]`
12	$[X,\ X-Y,\ -Z]$	$\left[x,\ -x,\ \frac{2}{3}\right]$	`[12]`

* Wyckoff site: `6j`, site symmetry: `..2`

Table 44: Wyckoff bond: `6a@6j`, bond symmetry: `..2`

No.	vector	center	mapping
1	$[X,\ 0,\ Z]$	$\left[x,\ 2x,\ \frac{1}{2}\right]$	`[1,-11]`
2	$[0,\ X,\ Z]$	$\left[-2x,\ -x,\ \frac{5}{6}\right]$	`[2,-10]`
3	$[-X,\ -X,\ Z]$	$\left[x,\ -x,\ \frac{1}{6}\right]$	`[3,-12]`
4	$[-X,\ 0,\ Z]$	$\left[-x,\ -2x,\ \frac{1}{2}\right]$	`[4,-8]`
5	$[0,\ -X,\ Z]$	$\left[2x,\ x,\ \frac{5}{6}\right]$	`[5,-7]`
6	$[X,\ X,\ Z]$	$\left[-x,\ x,\ \frac{1}{6}\right]$	`[6,-9]`

Table 45: Wyckoff bond: `6b@6j`, bond symmetry: `..2`

No.	vector	center	mapping
1	$[X,\ 2X,\ 0]$	$\left[x,\ 2x,\ \frac{1}{2}\right]$	`[1,11]`
2	$[-2X,\ -X,\ 0]$	$\left[-2x,\ -x,\ \frac{5}{6}\right]$	`[2,10]`
3	$[X,\ -X,\ 0]$	$\left[x,\ -x,\ \frac{1}{6}\right]$	`[3,12]`
4	$[-X,\ -2X,\ 0]$	$\left[-x,\ -2x,\ \frac{1}{2}\right]$	`[4,8]`
5	$[2X,\ X,\ 0]$	$\left[2x,\ x,\ \frac{5}{6}\right]$	`[5,7]`
6	$[-X,\ X,\ 0]$	$\left[-x,\ x,\ \frac{1}{6}\right]$	`[6,9]`

Table 46: Wyckoff bond: `12c@6j`, bond symmetry: `1`

No.	vector	center	mapping
1	$[X,\ Y,\ Z]$	$\left[x,\ 2x,\ \frac{1}{2}\right]$	`[1]`
2	$[-Y,\ X-Y,\ Z]$	$\left[-2x,\ -x,\ \frac{5}{6}\right]$	`[2]`
3	$[-X+Y,\ -X,\ Z]$	$\left[x,\ -x,\ \frac{1}{6}\right]$	`[3]`
4	$[-X,\ -Y,\ Z]$	$\left[-x,\ -2x,\ \frac{1}{2}\right]$	`[4]`
5	$[Y,\ -X+Y,\ Z]$	$\left[2x,\ x,\ \frac{5}{6}\right]$	`[5]`
6	$[X-Y,\ X,\ Z]$	$\left[-x,\ x,\ \frac{1}{6}\right]$	`[6]`
7	$[Y,\ X,\ -Z]$	$\left[2x,\ x,\ \frac{5}{6}\right]$	`[7]`
8	$[X-Y,\ -Y,\ -Z]$	$\left[-x,\ -2x,\ \frac{1}{2}\right]$	`[8]`
9	$[-X,\ -X+Y,\ -Z]$	$\left[-x,\ x,\ \frac{1}{6}\right]$	`[9]`
10	$[-Y,\ -X,\ -Z]$	$\left[-2x,\ -x,\ \frac{5}{6}\right]$	`[10]`
11	$[-X+Y,\ Y,\ -Z]$	$\left[x,\ 2x,\ \frac{1}{2}\right]$	`[11]`
12	$[X,\ X-Y,\ -Z]$	$\left[x,\ -x,\ \frac{1}{6}\right]$	`[12]`

* Wyckoff site: `12k`, site symmetry: `1`

Table 47: Wyckoff bond: `12a@12k`, bond symmetry: `1`

No.	vector	center	mapping
1	$[X,\ Y,\ Z]$	$[x,\ y,\ z]$	`[1]`
2	$[-Y,\ X-Y,\ Z]$	$\left[-y,\ x-y,\ z+\frac{1}{3}\right]$	`[2]`
3	$[-X+Y,\ -X,\ Z]$	$\left[-x+y,\ -x,\ z+\frac{2}{3}\right]$	`[3]`
4	$[-X,\ -Y,\ Z]$	$[-x,\ -y,\ z]$	`[4]`
5	$[Y,\ -X+Y,\ Z]$	$\left[y,\ -x+y,\ z+\frac{1}{3}\right]$	`[5]`
6	$[X-Y,\ X,\ Z]$	$\left[x-y,\ x,\ z+\frac{2}{3}\right]$	`[6]`
7	$[Y,\ X,\ -Z]$	$\left[y,\ x,\ \frac{1}{3}-z\right]$	`[7]`
8	$[X-Y,\ -Y,\ -Z]$	$[x-y,\ -y,\ -z]$	`[8]`
9	$[-X,\ -X+Y,\ -Z]$	$\left[-x,\ -x+y,\ \frac{2}{3}-z\right]$	`[9]`
10	$[-Y,\ -X,\ -Z]$	$\left[-y,\ -x,\ \frac{1}{3}-z\right]$	`[10]`
11	$[-X+Y,\ Y,\ -Z]$	$[-x+y,\ y,\ -z]$	`[11]`
12	$[X,\ X-Y,\ -Z]$	$\left[x,\ x-y,\ \frac{2}{3}-z\right]$	`[12]`